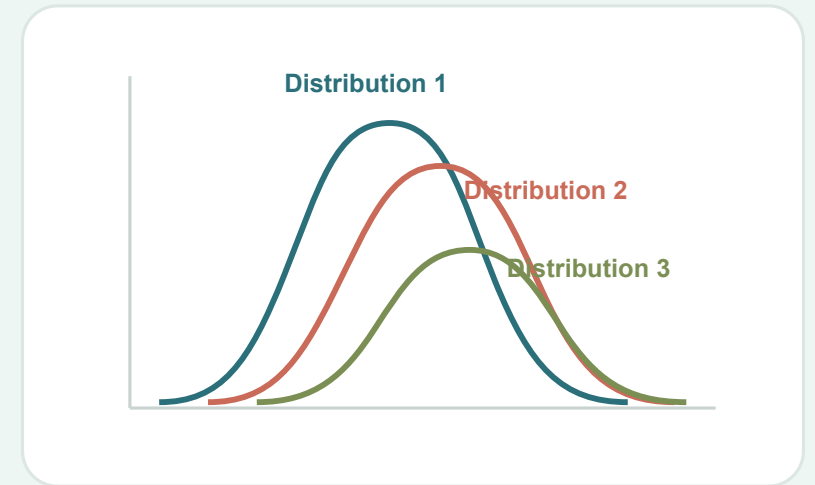


# Many distributions at once

Exploring several distributions together

Based on Chapter 7 of *Fundamentals of Data Visualization*

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TODAY

## Twelve months, twelve distributions

- Last week, one curve described every passenger's age.
- Today each month is its own distribution: 29, 30, or 31 temperatures, one per day.
- The question is how to compare all twelve without hiding them.

The data are the daily mean temperature in Lincoln, Nebraska, for every day of 2016. There are 366 days, because 2016 was a leap year. Temperature is in degrees Fahrenheit.

## PART 1

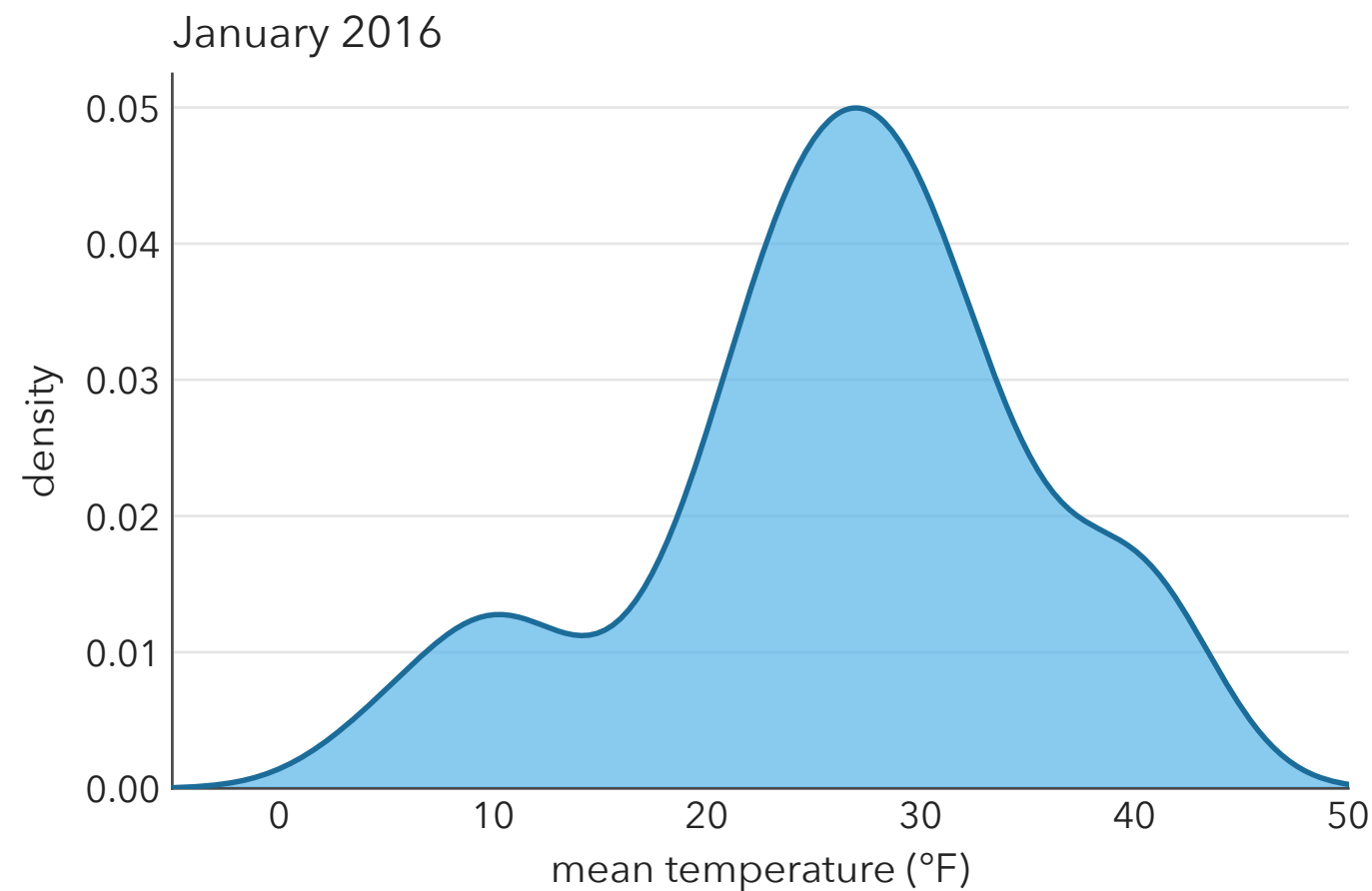
# One month is easy. Twelve are not.

A density still works for a single month. The trouble starts when the months have to share one picture.

# January, by itself

January has 31 days. The coldest mean was 5°F and the warmest was 41°F. These are the first six days.

| DATE  | MEAN °F |
|-------|---------|
| Jan 1 | 24      |
| Jan 2 | 23      |
| Jan 3 | 23      |
| Jan 4 | 17      |
| Jan 5 | 29      |
| Jan 6 | 33      |



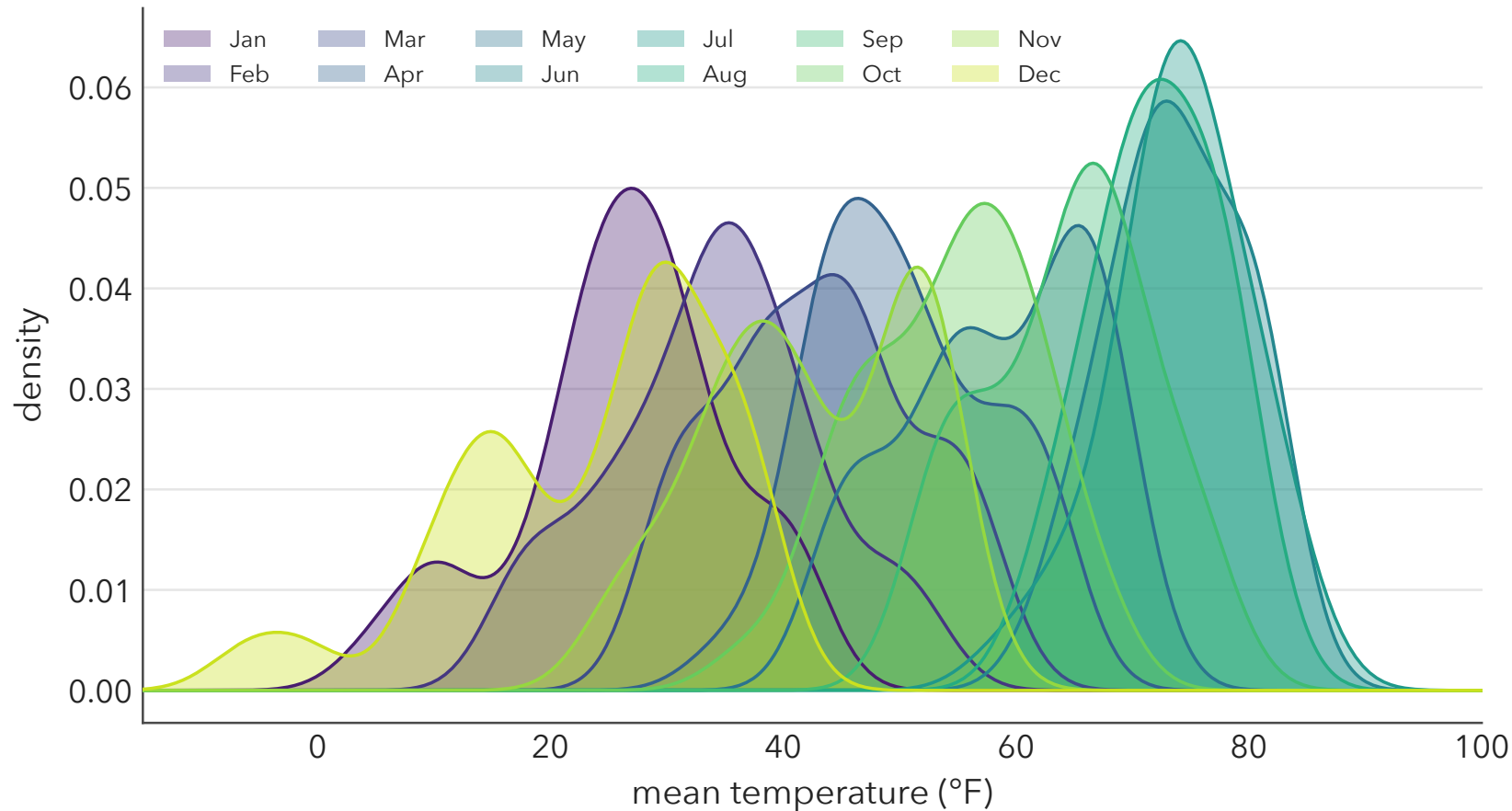
## QUESTION

**Will June sit to the left of January, or to the right?**

Same horizontal axis as the January curve. June is hotter. Where does its density go?

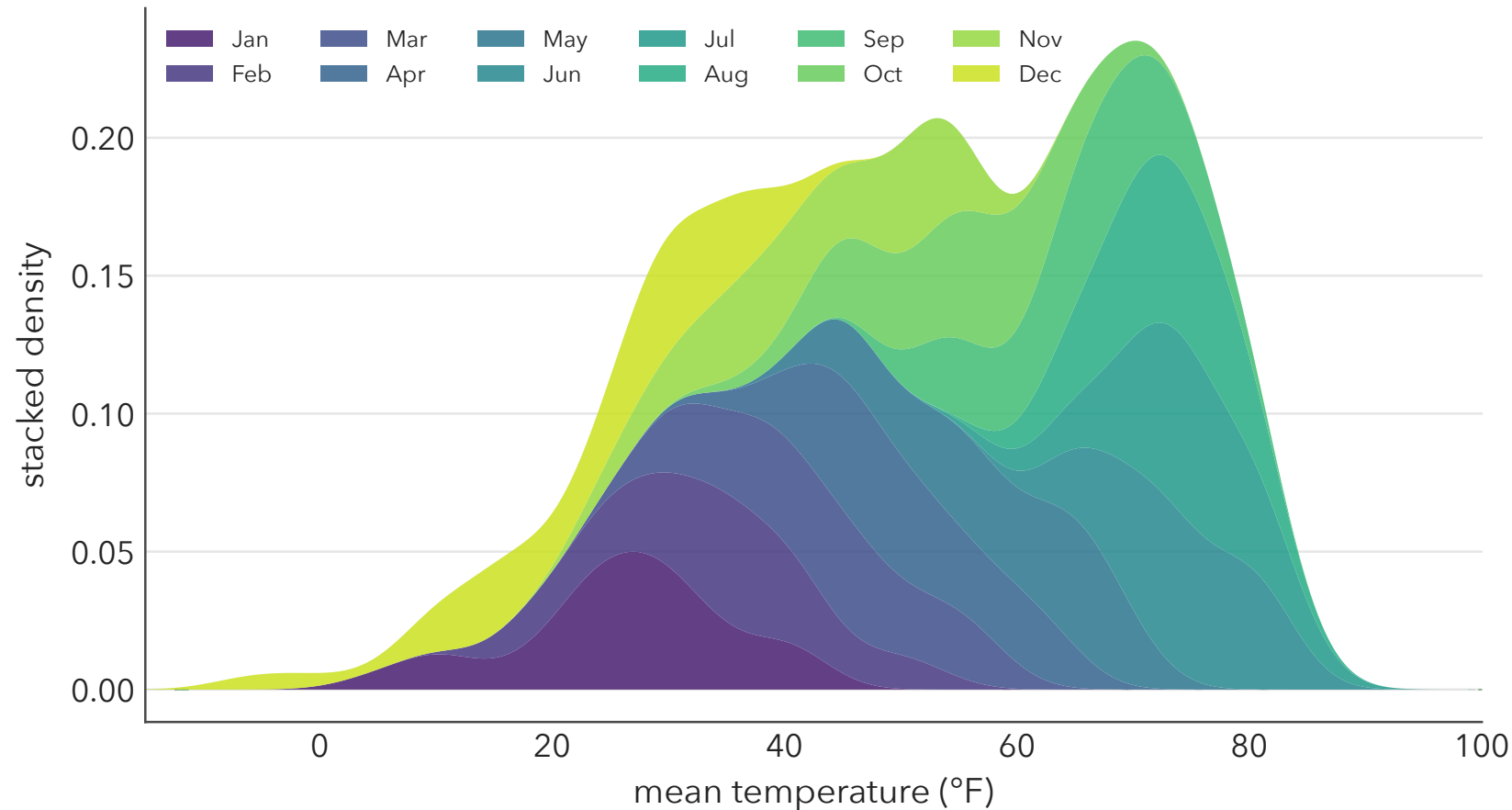
# Overlapping the twelve curves hides them

To the right. Each month is still a density, drawn on the same axes. Twelve translucent shapes cover one another. You can see that summer sits to the right of winter, and almost nothing else.



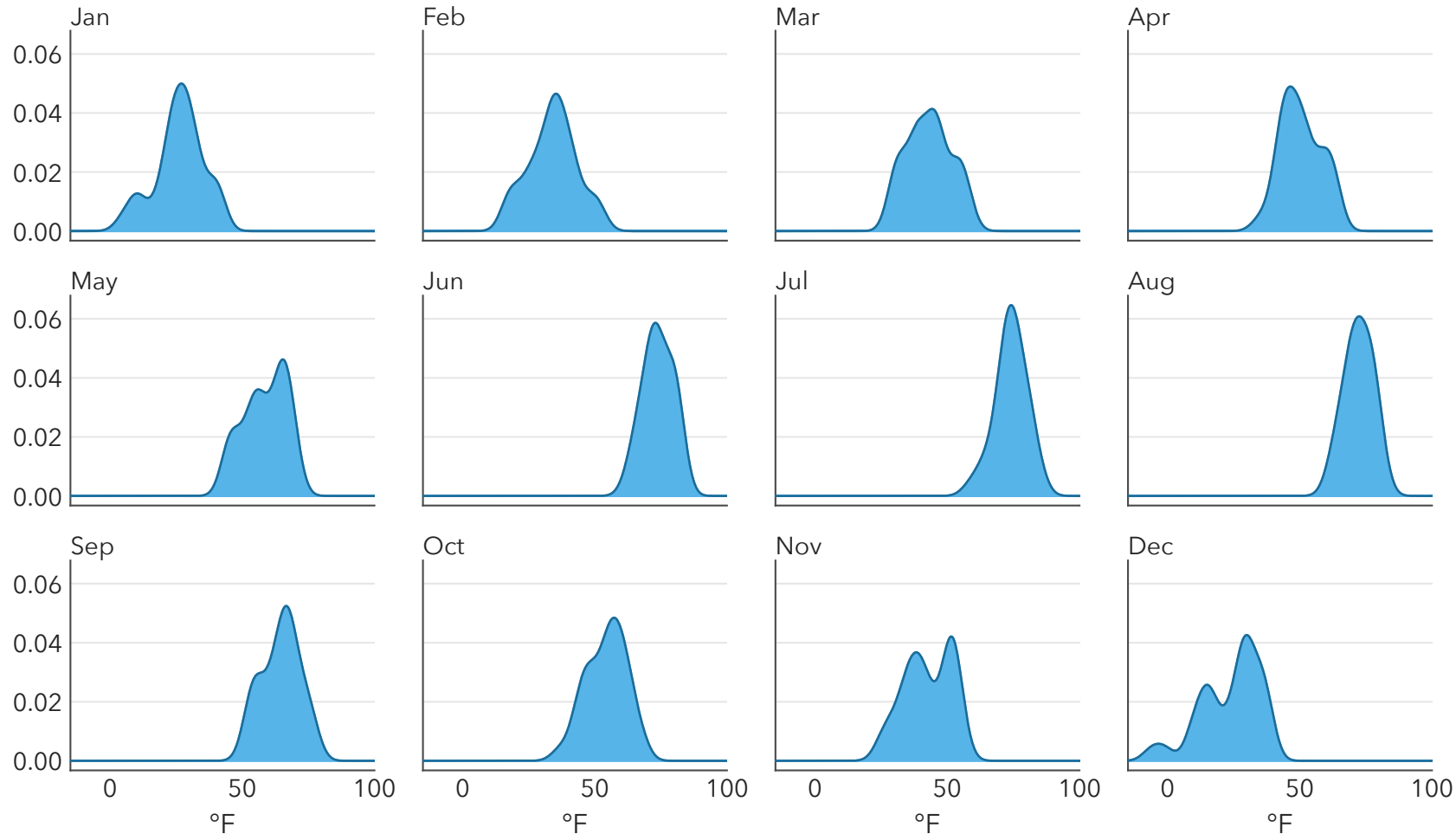
# Stacking them hides them too

Stacking sets each month on top of the others, so a month's baseline moves. The shape you see is no longer that month's density. This is rarely a good choice, for the same reason a stacked histogram was hard to read last week.



# Small multiples are clearer, and still hard to compare

Each month gets its own panel, with the same horizontal scale. You can read one month. Comparing the peak of March with the peak of October means jumping between panels.



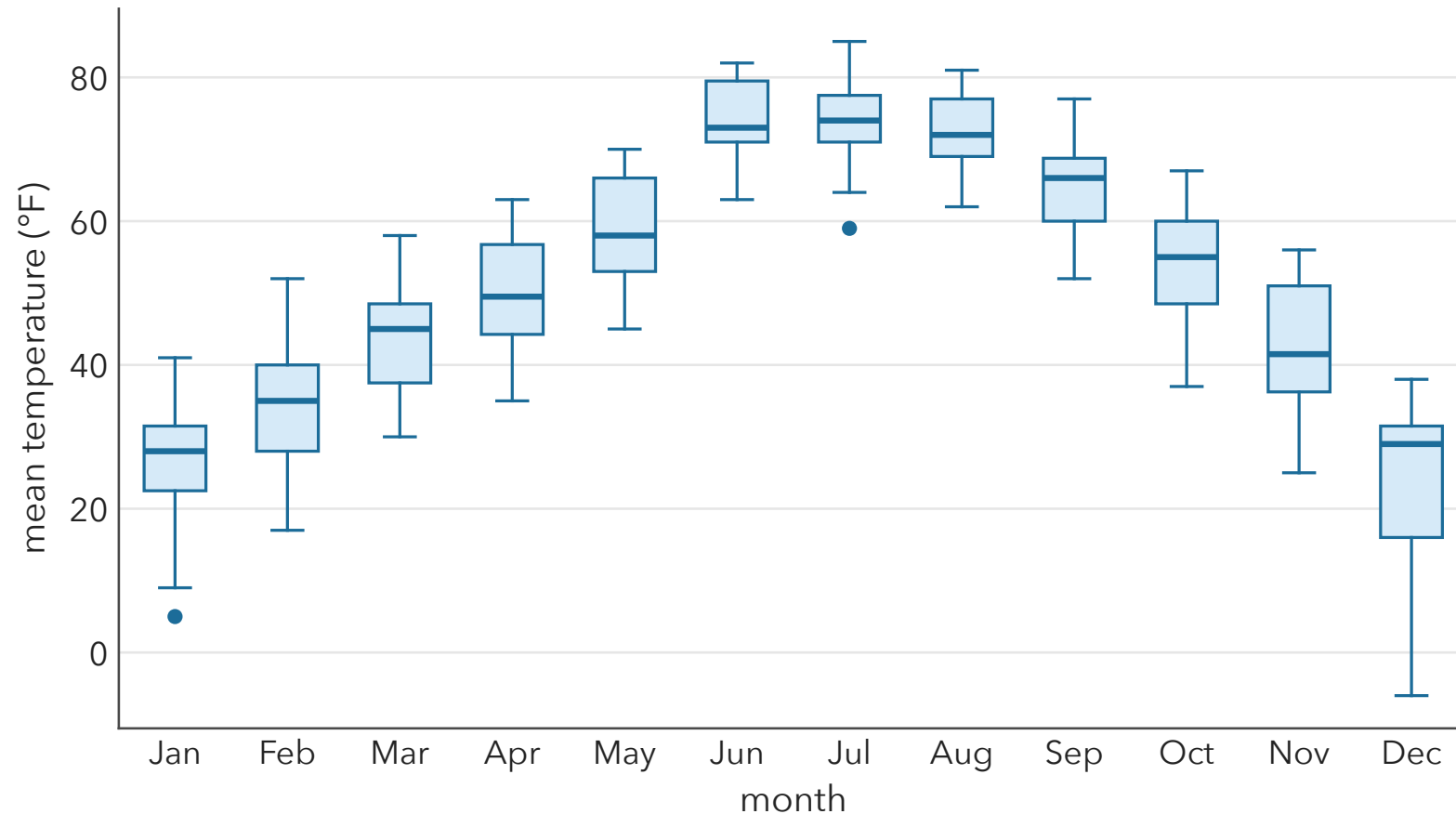
## PART 2

# Put the month on the horizontal axis

Temperature stays vertical. The month becomes a position, so the eye can move from January to December in one line.

# A boxplot is a compact summary of one month

Summer sits high and the boxes are short, so those days are hot and similar to each other. Winter sits low and December's box is tall, so those days vary. January's coldest day is 5°F, and that day is the dot under January.



## QUESTION

# Should we delete January's 5°F day?

It is drawn as a dot, apart from the box. Is that day a mistake in the data?

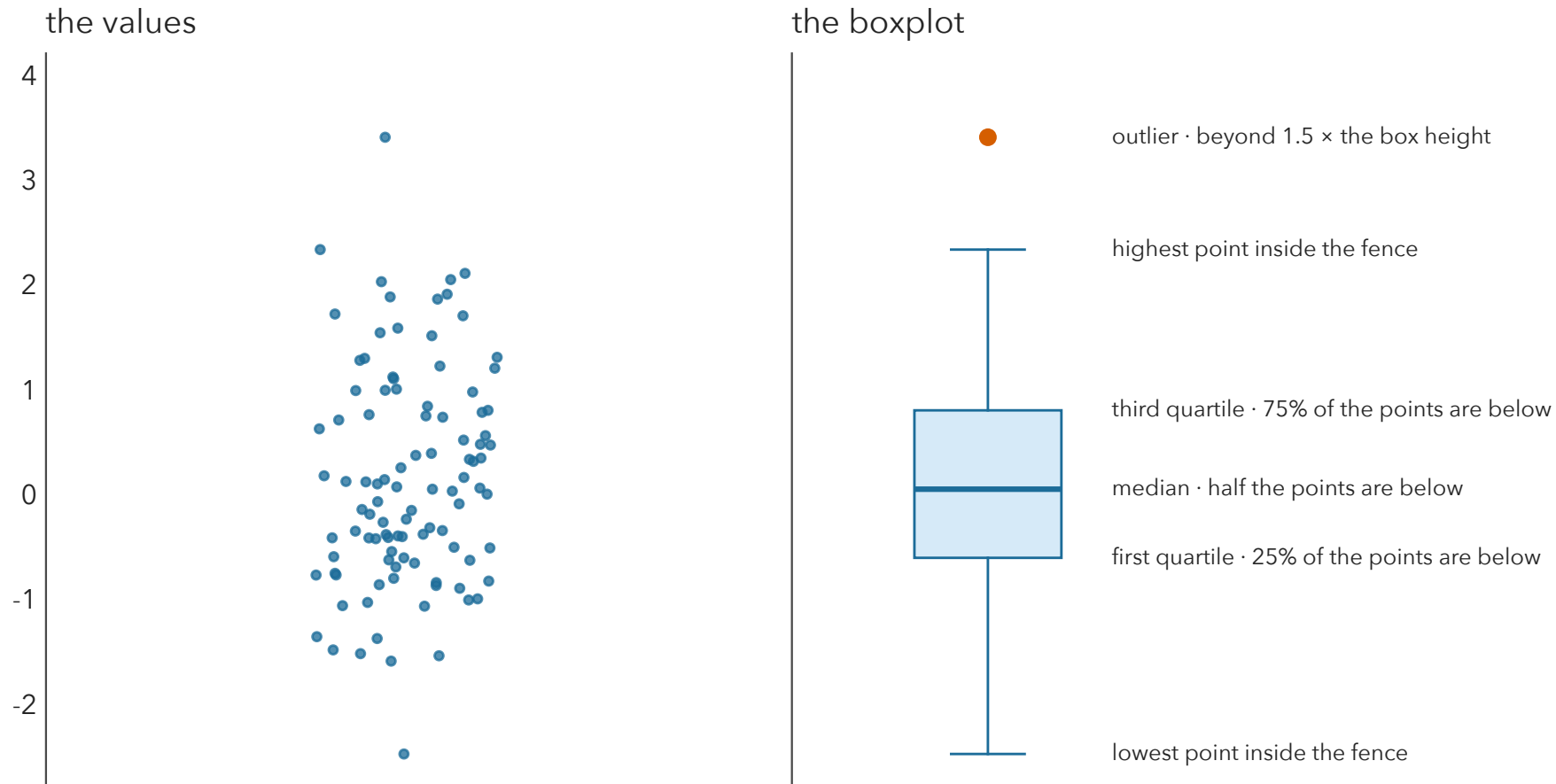
## **No. That day stays.**

The dot is a real day in January 2016. It is colder than the middle half of the month, so the boxplot draws it by itself. A dot marks a day. It does not throw the day out.

The next picture is a made-up sample, used only to name the parts of a box. Those numbers are not Lincoln temperatures.

# How to read a boxplot

This sample is made up, so the labels have room. The box holds the middle half of the values. The line in the box is the median. The whiskers stop at the last value inside the fence, and a point beyond the fence is drawn by itself.



## QUESTION

# About where is the upper fence?

In the made-up sample, the box runs from  $-0.6$  to  $0.8$ . The fence sits 1.5 box-heights past the top of the box. Is the point at  $3.4$  inside that fence?

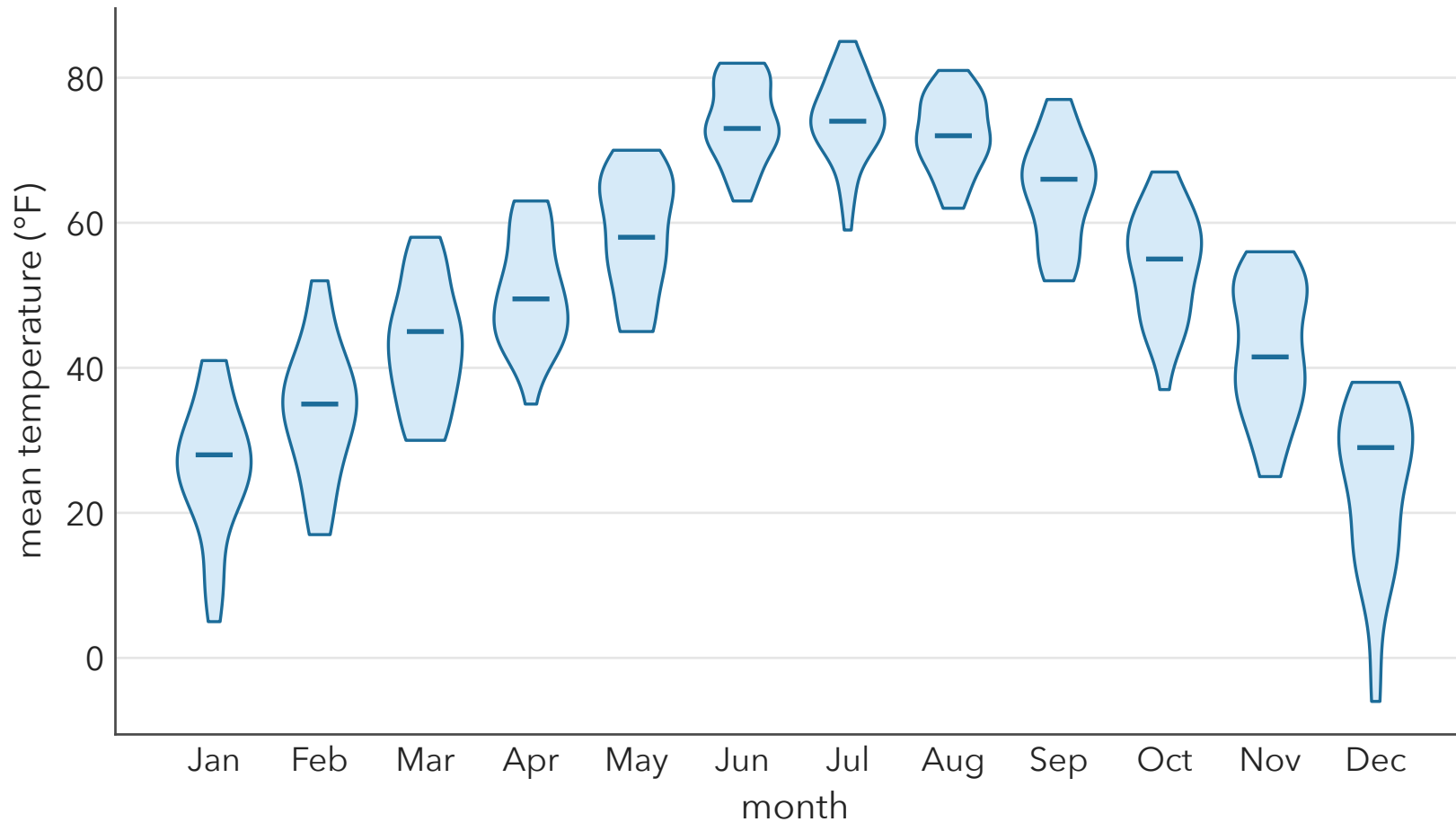
## The fence is 1.5 boxes past the quartiles

About 2.9, and 3.4 is outside it, so that point is the dot. The height of the box is  $0.8 - (-0.6) = 1.4$ . One and a half boxes is about 2.1. Add that to 0.8 and the upper fence is about 2.9.

A boxplot does not show the shape. Two months can share a box and a median and still pile up in different places. That is what the violin is for.

# A violin is a density, turned sideways and mirrored

The widest part of a month is where its days are most common. Summer is a short, high shape. Spring and fall are taller, because a month can hold both a cold week and a warm one.



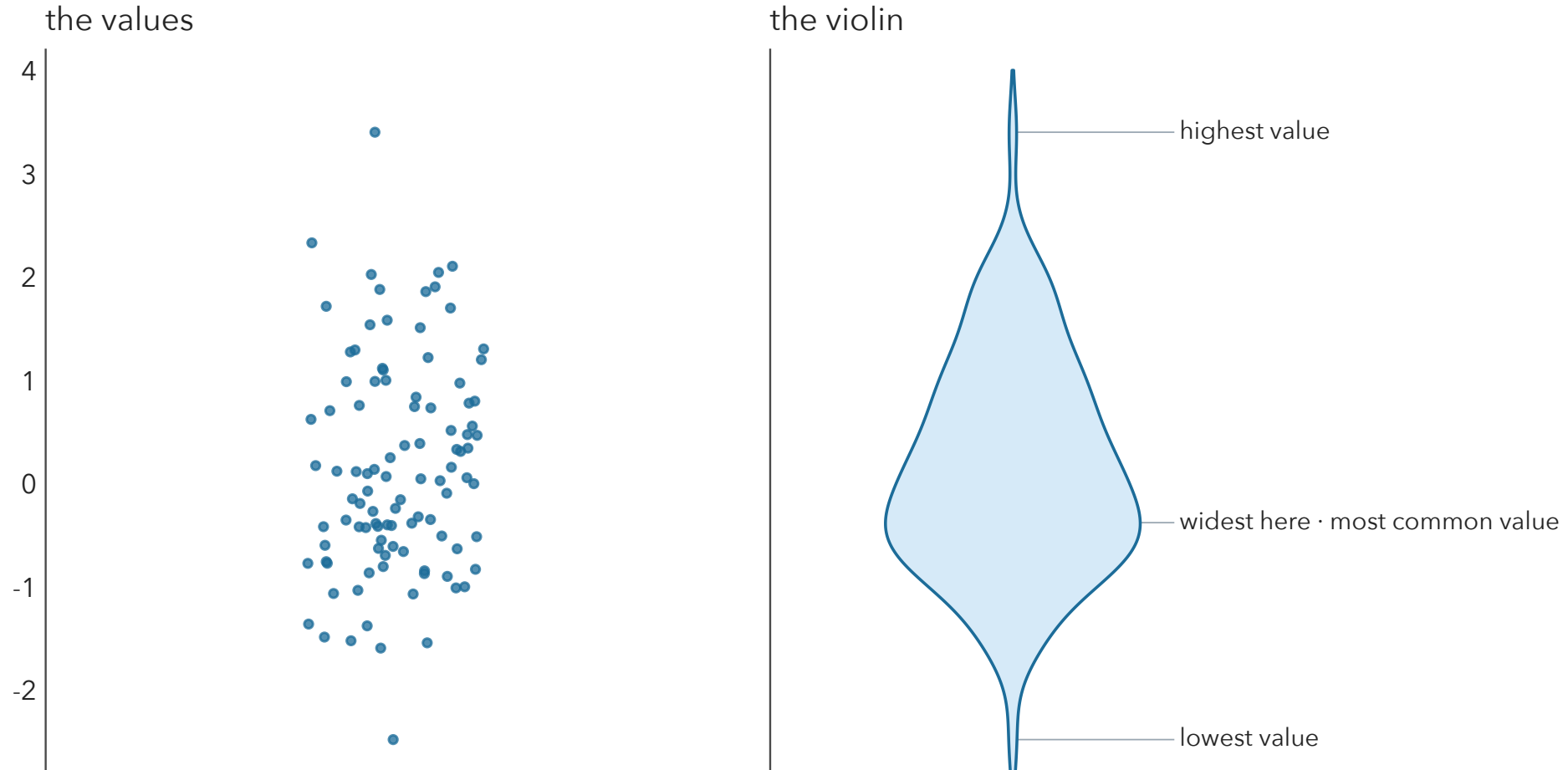
## QUESTION

# Does the mirror mean two groups?

Each violin is drawn to the left and to the right of its month. Is that two groups of days?

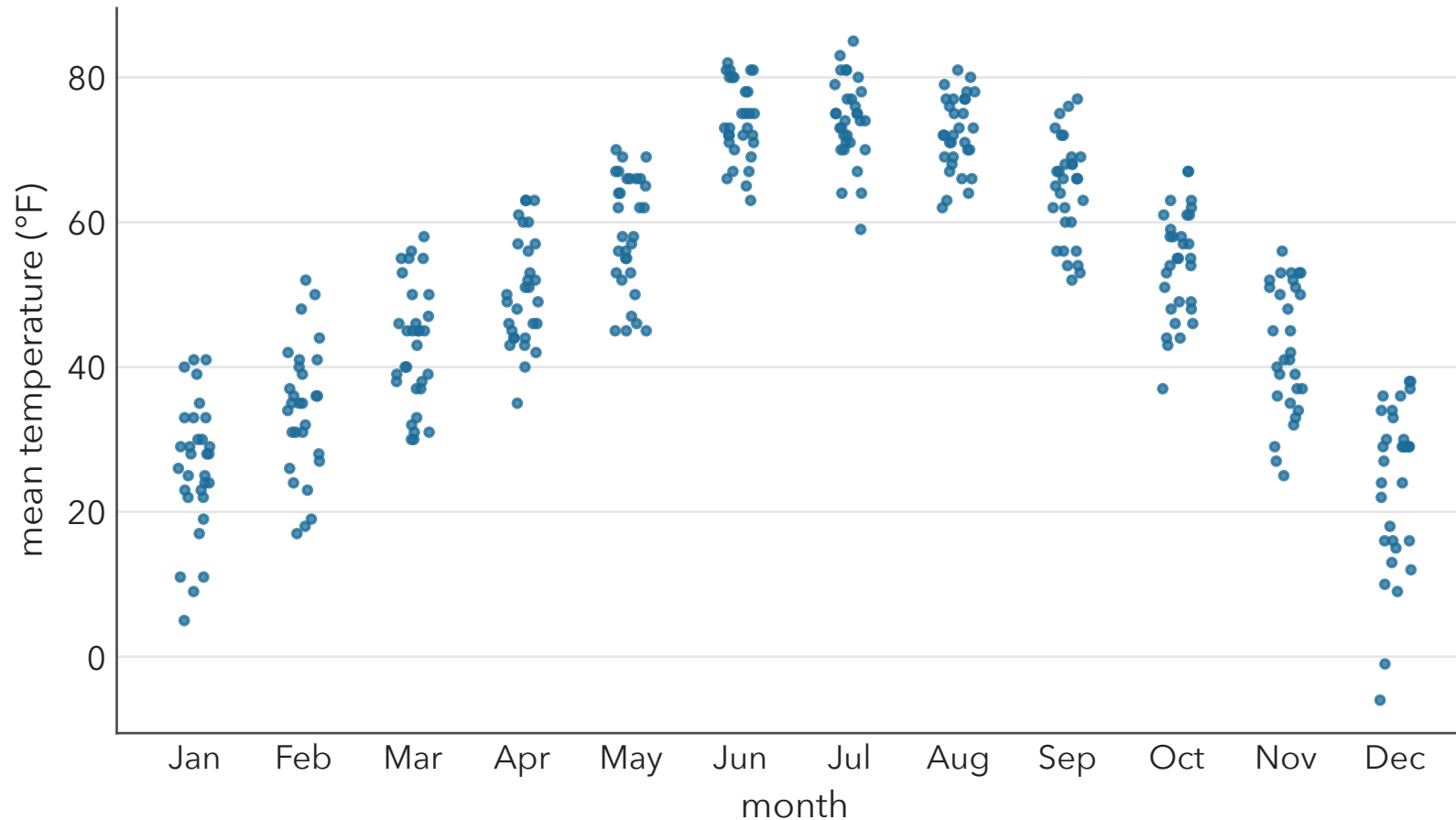
# How to read a violin

No. Same made-up sample as the boxplot. The width at a height is the density there. The mirror is only so the shape is easy to see. It does not mean there are two groups.



# A strip chart shows every day

Each dot is one day. The dots are nudged left and right, at random, so days with the same temperature do not land on one spot. The nudge is horizontal only. A day's temperature does not change.



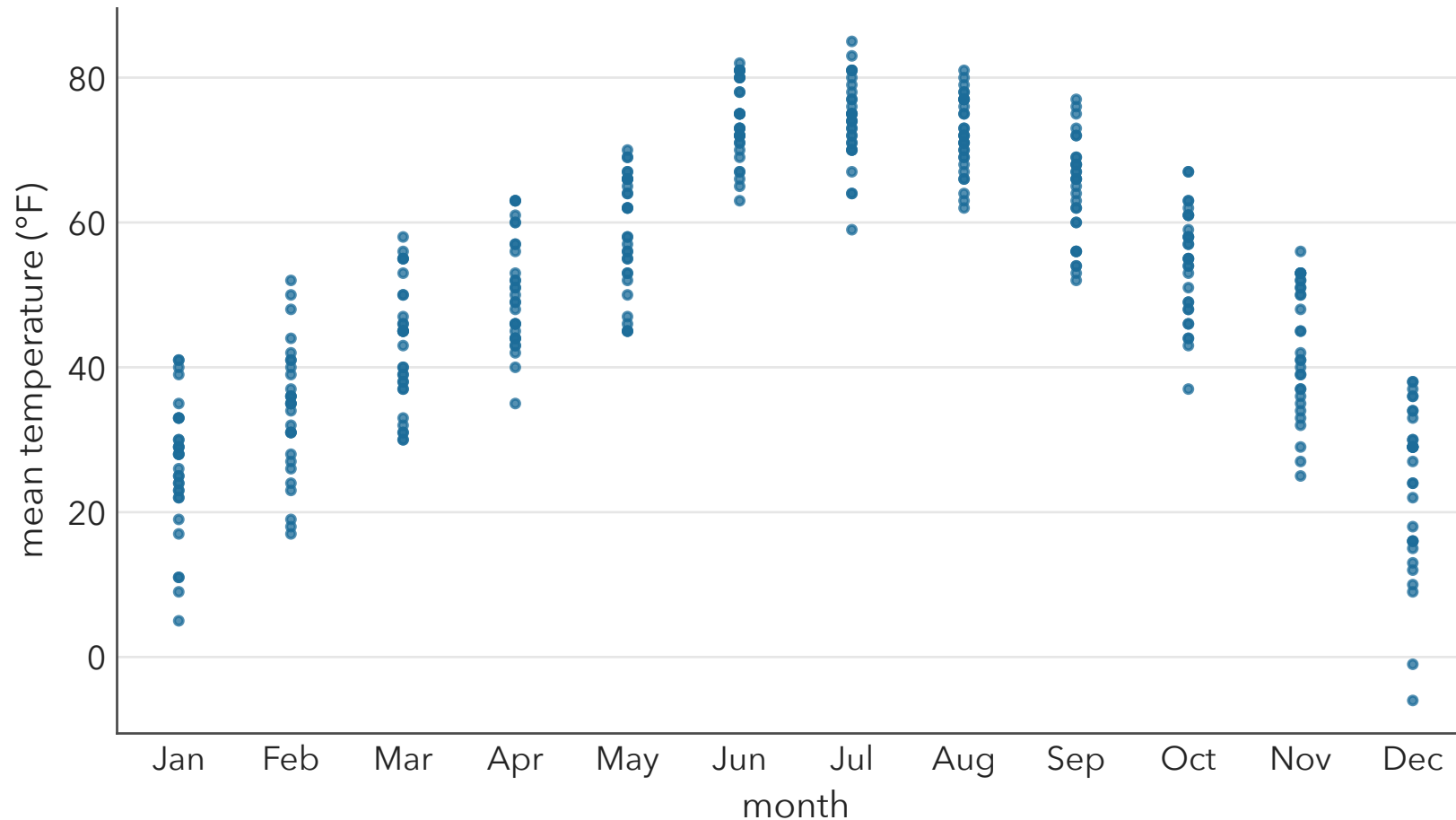
## QUESTION

# What if the sideways nudge is removed?

Two days in the same month have the same temperature. Where do those two dots go?

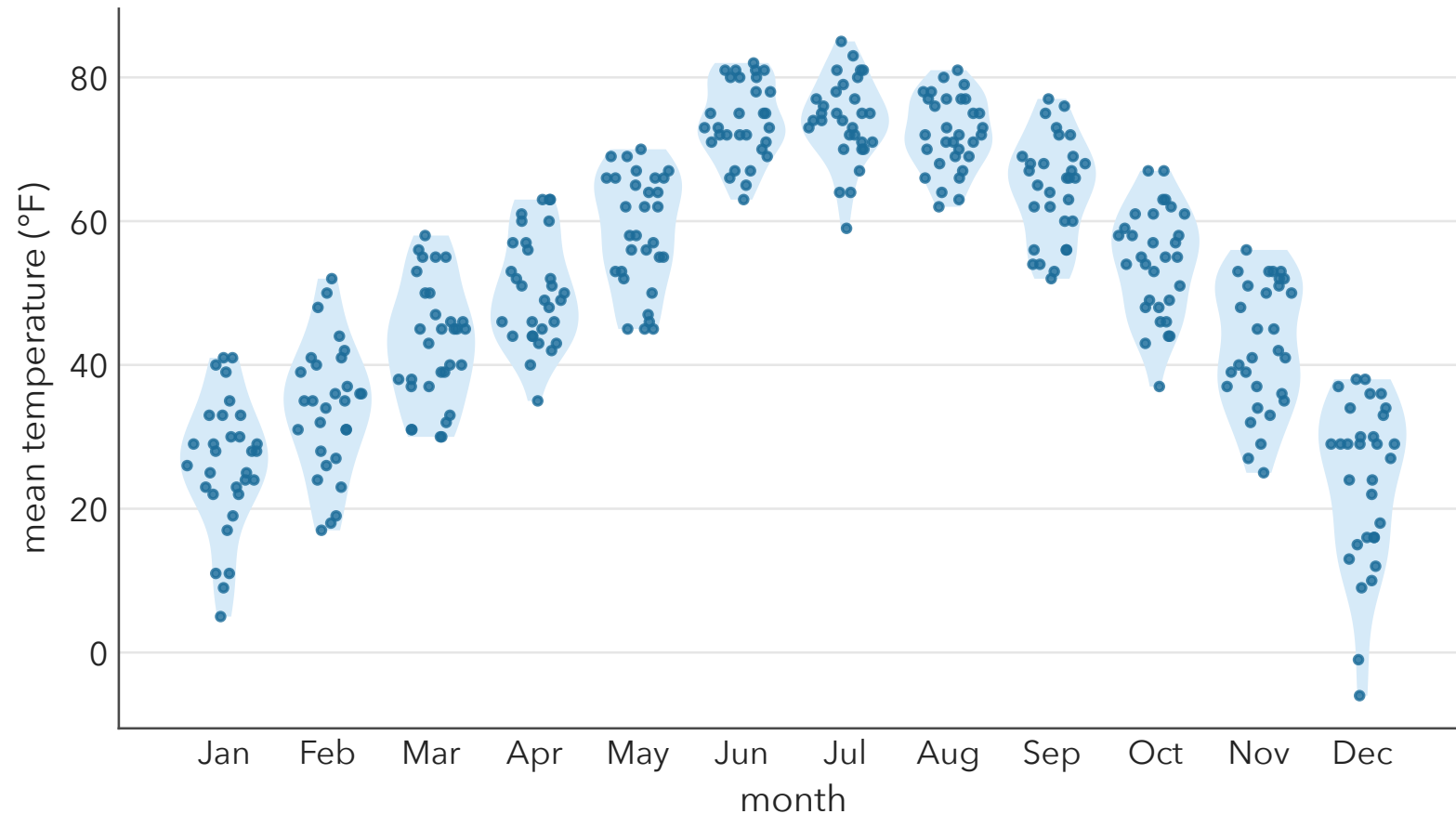
# Without the nudge, days sit on top of each other

On the same spot. Same temperatures, no horizontal shift. A month that had many days at 70°F looks like a single dot. The picture understates how common that temperature was.



# A sina plot lets the dots follow the violin

The pale shape is the violin. Each day is still a dot, and the sideways nudge is only as wide as the violin at that temperature. Where the month is common, the dots spread out. Where it is rare, they stay near the center. The plot is named for Sina Hadi Sohi.



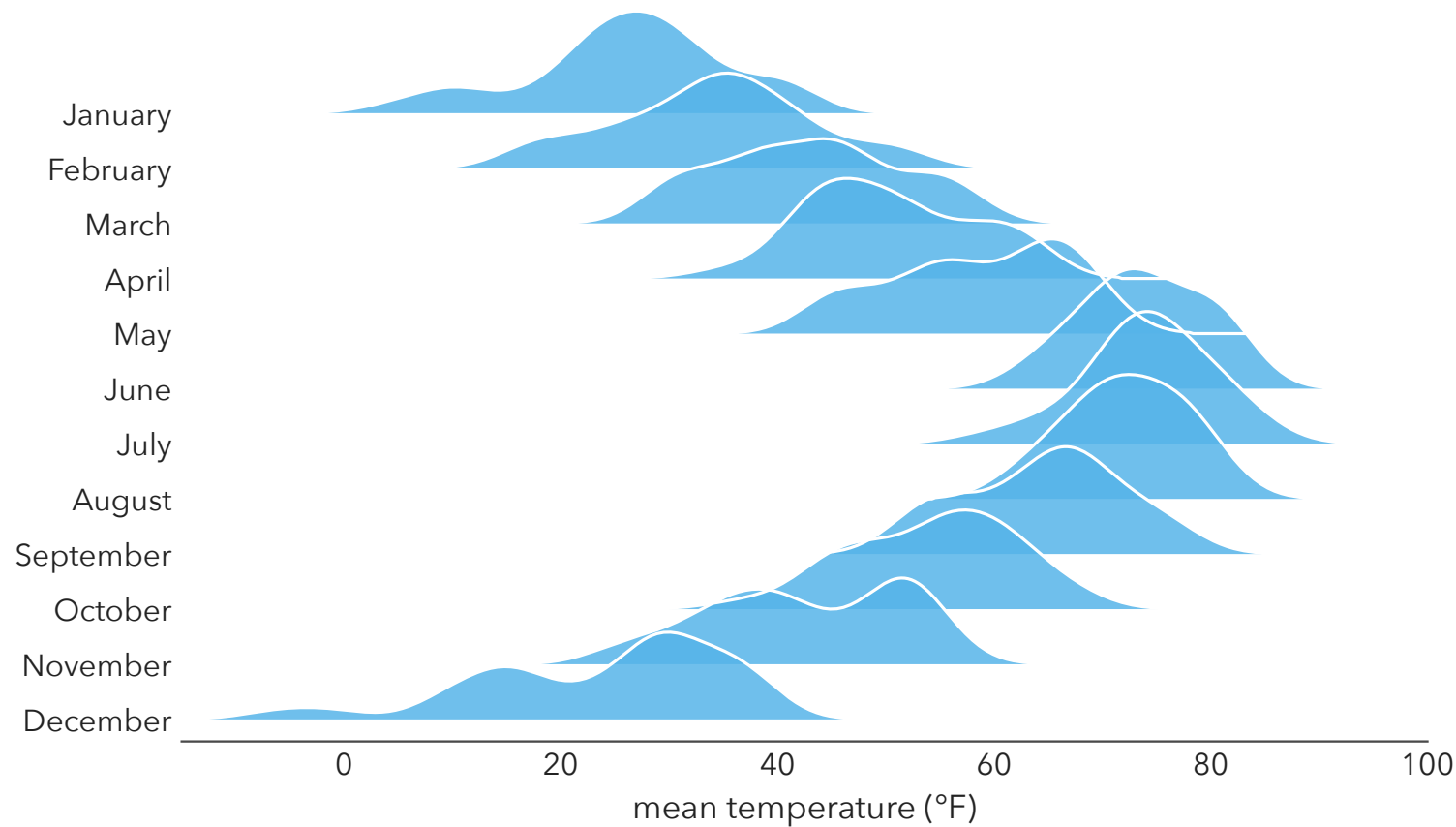
### PART 3

# Give each density its own row

A ridgeline is last week's density, repeated once per month and lifted so the curves no longer cover each other.

# Ridgelines keep the shape and the order of the year

January is the top row. One color is enough, because the month is already named on the left. A second color would only compete with the shape. The curves overlap a little, on purpose, so the eye can follow the year from cold to hot and back.

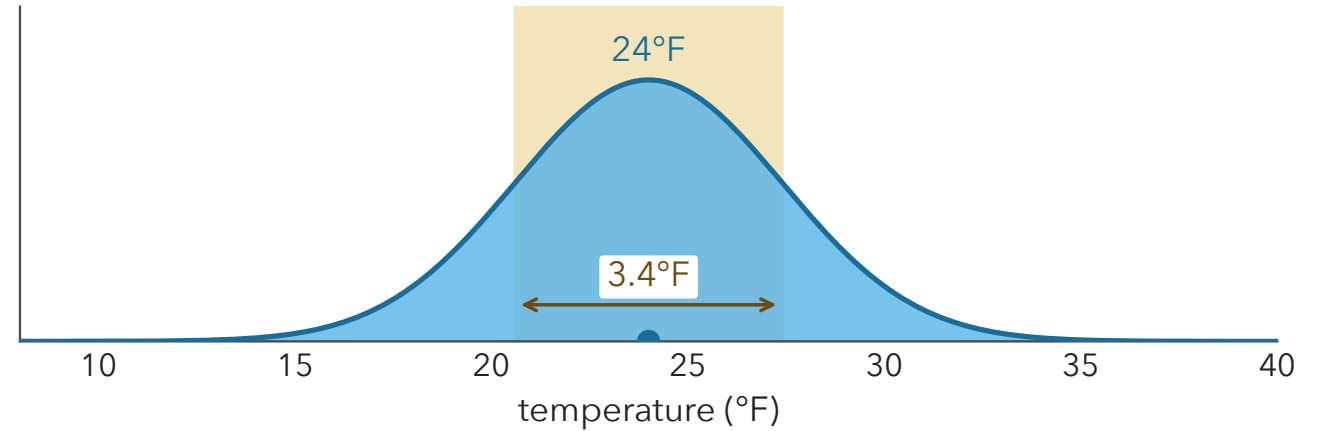


# Bandwidth is how wide each day's bell is

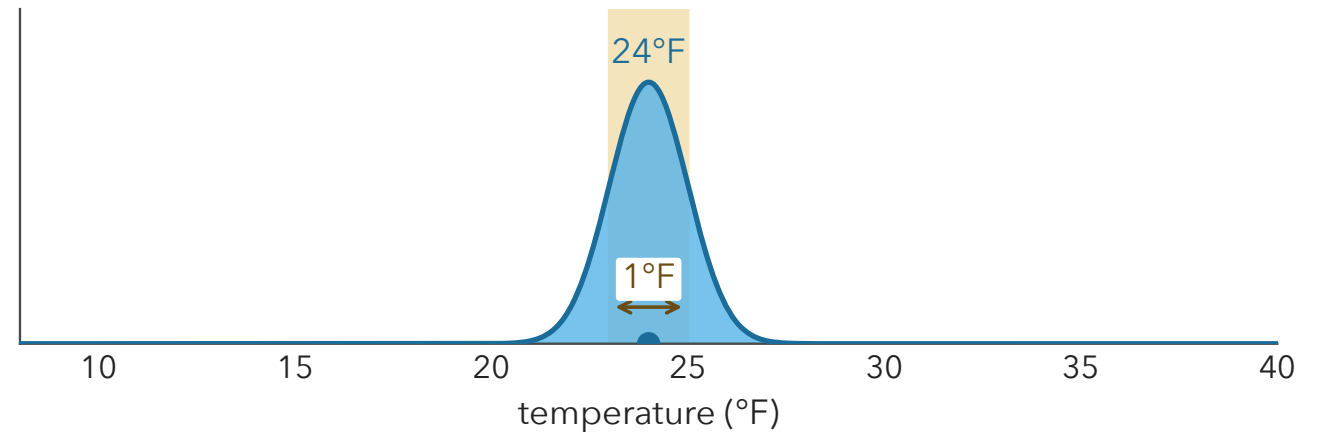
A density puts one bell on every day, then adds the bells. Bandwidth is the width of that bell, in degrees Fahrenheit. Last week the width was in years of age. Here it is in degrees.

January 1 was 24°F. The gold band is one bandwidth on either side of that day. Both bells are that one day. Only the width changes.

Bandwidth 3.4°F



Bandwidth 1°F



## QUESTION

# What if the bandwidth drops to 1°F?

The ridgelines use 3.4°F, so a day's bell reaches a few degrees on either side of that day. At 1°F it stays close to the day itself. What will a month's curve do?

## **The curve follows the jumps**

Each bell is only about 1 degree wide, so the curve traces the small rises and falls in a month of about 30 days. At 3.4°F the bells overlap their neighbors and the month becomes one smooth shape. The worksheet asks you to draw both.

## PART 4

# The same pictures in matplotlib

pandas holds the table. matplotlib draws the boxes, the violins, and the dots.

# Read the table

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

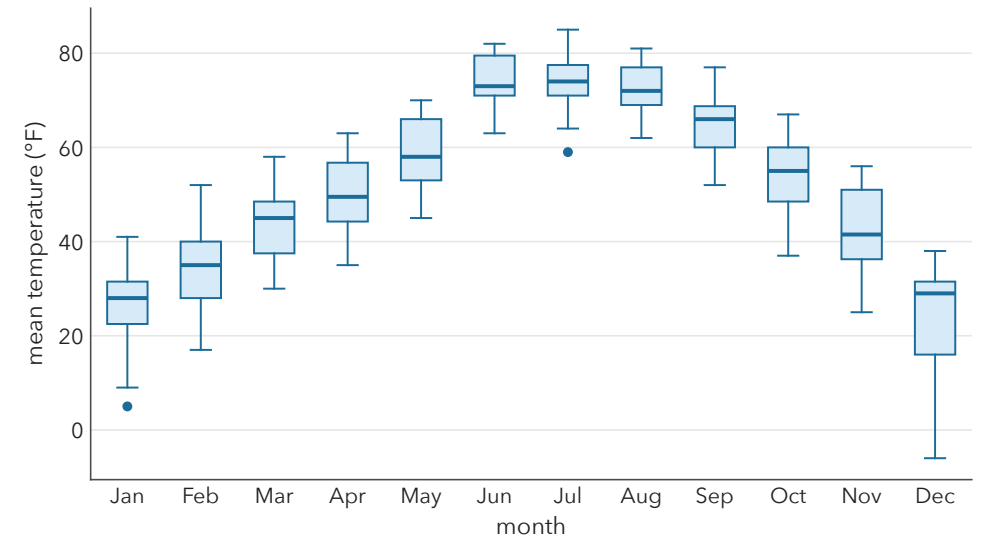
temps = pd.read_csv("lincoln_temps.csv")
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
```

One row is one day. `mean_temp` is that day's average temperature, in °F. `month` is the three-letter name.

The list `months` is the order of the year. Without it, a plot that sorts the names would put April before August.

# A boxplot takes one list of temperatures per month

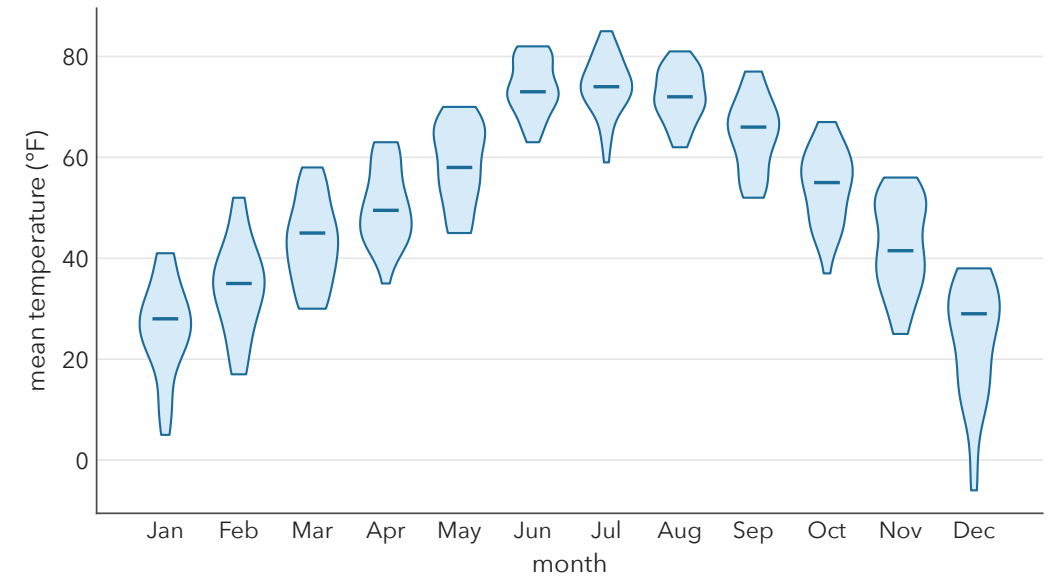
```
groups = [  
    temps.loc[temps["month"] == month, "mean_temp"]  
    for month in months  
]  
fig, ax = plt.subplots()  
ax.boxplot(groups)  
ax.set_xticks(range(1, 13))  
ax.set_xticklabels(months)  
ax.set_ylabel("mean temperature (°F)")
```



`loc` keeps the rows whose month matches. The result is twelve lists, in the order of `months`.

# A violin uses the same twelve lists

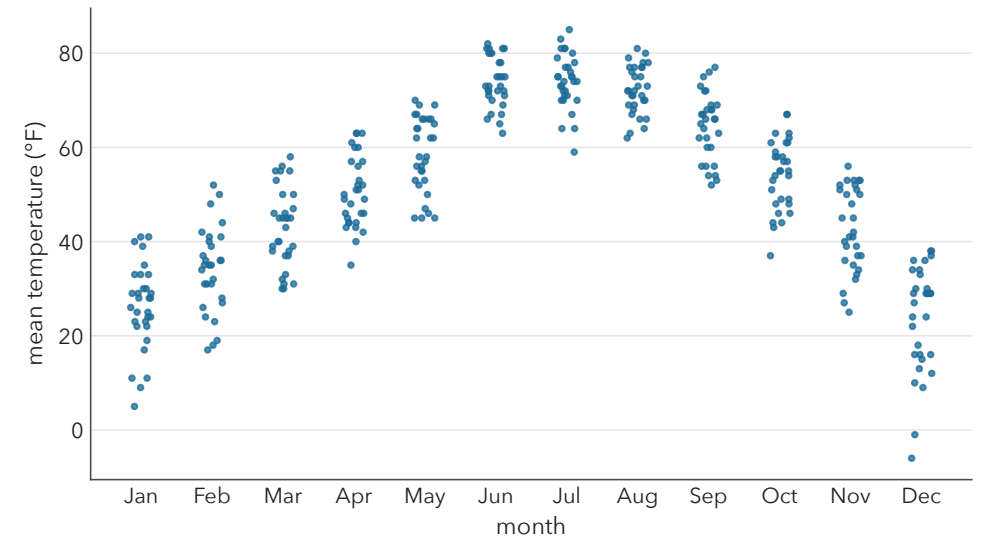
```
fig, ax = plt.subplots()
ax.violinplot(groups, showmedians=True)
ax.set_xticks(range(1, 13))
ax.set_xticklabels(months)
ax.set_ylabel("mean temperature (°F)")
```



`violinplot` numbers the groups 1, 2, 3, and so on. `set_xticks` marks the positions 1 through 12. `set_xticklabels` writes the month names on those marks.

# Jitter moves a dot sideways, not up

```
rng = np.random.default_rng(320)
for i, month in enumerate(months, start=1):
    y = temps.loc[temps["month"] == month, "mean_temp"]
    x = i + rng.uniform(-0.15, 0.15, size=len(y))
    ax.scatter(x, y, s=8)
```



`uniform` draws a random shift between  $-0.15$  and  $0.15$ . The seed 320 makes that shift the same every time you run the cell.  
`enumerate` counts from 1 so the first month sits at  $x = 1$ .

# A case for each picture

| PICTURE     | A CASE WHERE IT HELPS                                                                                                                                                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Boxplot     | Exam scores for 12 sections. The boxes show which section sits higher, and a dot marks a score far from its own section. The shape of the scores stays hidden.                          |
| Violin      | Commute times that pile up twice, near 20 minutes and again near 50. The violin is wide in both places. A boxplot of the same times would hide the two piles.                           |
| Strip chart | Quiz scores for a class of 25. One dot per student. A sideways nudge keeps two equal scores from landing on the same spot.                                                              |
| Sina        | Those same 25 scores, when the shape matters too. The dots spread out where scores are common, and stay near the center where a score is rare.                                          |
| Ridgeline   | Daily steps for each month of a year, January at the top. The months are in order, so the eye can follow the year. One color is enough, because the month is already named on the left. |

## What the pictures were for

- **One panel each.** Overlapping and stacked densities hide the months. Small multiples are readable, and hard to compare.
- **Boxplot.** The box is the middle half. The line is the median. A dot is a day beyond the fence, 1.5 boxes past the quartile.
- **Violin.** A density on its side, then mirrored. The widest point is the most common temperature.
- **Strip chart.** One dot per day. Nudge sideways so equal temperatures do not stack. Do not nudge the temperature itself.
- **Ridgeline.** One density per month, each on its own row, in one color. Bandwidth is the width of each day's bell, in degrees.

Further reading

## The chapter, and the drawing functions

- Claus O. Wilke, [Chapter 7: Visualizing many distributions at once](#)
- matplotlib, [Axes.boxplot](#) and [Axes.violinplot](#)
- scikit-learn, [KernelDensity](#), used here for the ridgelines